

**AUTUMN
2023-2024**

CS426 Computer Graphics

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Time allowed: 2 hours

Answer at least **three** questions

Your mark will be based on your best **three** answers

All questions carry equal marks

Instructions

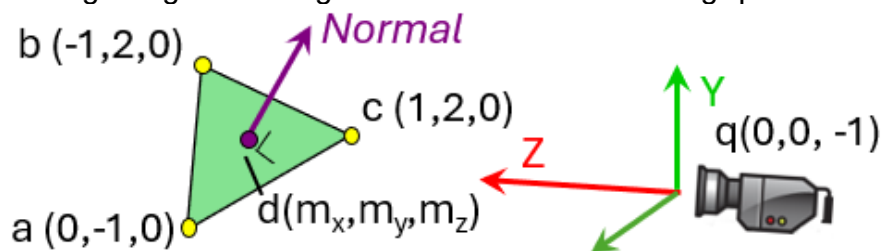
	Yes	No	N/A
Formulae and Tables book allowed (<i>i.e. available on request</i>)		X	
Formulae and Tables book required (<i>i.e. distributed prior to exam commencing</i>)		X	
Statistics Tables and Formulae allowed (<i>i.e. available on request</i>)		X	
Statistics Tables and Formulae required (<i>i.e. distributed prior to exam commencing</i>)		X	
Dictionary allowed (<i>supplied by the student</i>)		X	
Non-programmable calculator allowed	X		
Students required to write in and return the exam question paper		X	

[40 marks]

- 1 (a) Explain the term right-handed co-ordinate system. Include a diagram with your answer. Include an explanation of how to measure the direction of increasing angle. [10 marks]

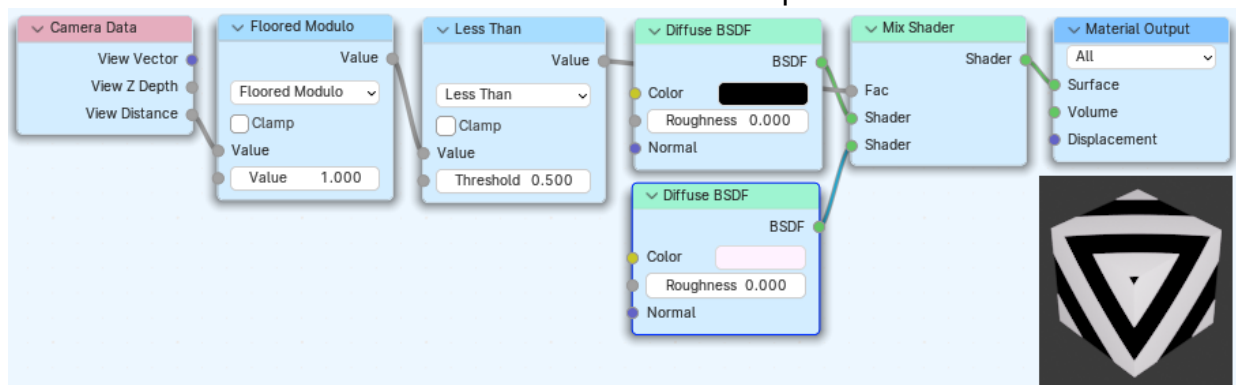
Explain the term raytracing. Name and describe four different types of rays generated when ray tracing. Include a diagram with your answer.

- (b) Consider the following figure showing the geometry of a camera viewing the green triangle and answer the following questions. [10 marks]



- Calculate the centroid, m , of the triangle.
- Calculate the surface normal to the triangle; you can assume a clockwise winding-order. Explain your reasoning.
- Calculate if the normal is pointing towards or away from the camera. Explain your reasoning.

- (c) The following Blender shader was used to texture the default cube. Consider the shader and answer the questions that follow. [10 marks]



- Explain the function of each of the nodes.
- Explain how the shader creates the effect seen.
- Sketch (or describe) the approximate position of the camera in relation to the cube.

Continued next page.

- (d) Explain how you would draw on paper a perspective view of a checkered floor. Include a labeled diagram with your answer. Label the vanishing point. [10 marks]

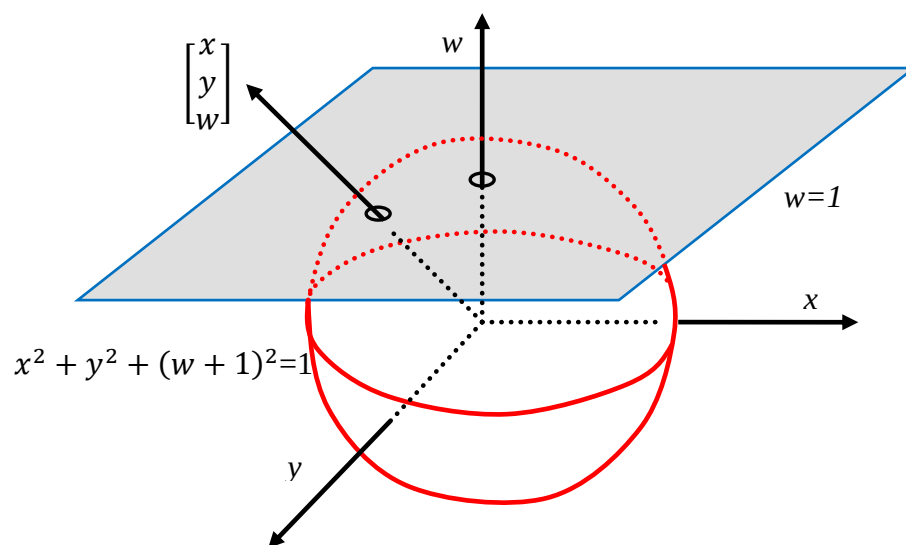
[40 marks]**2 (a)** Describe each of the following camera models.**[10 marks]**

- (i) First person
- (ii) Second person
- (iii) Third person
- (iv) Fly through
- (v) Arc Ball

(b) Explain the term homogeneous co-ordinate. Write the following vector in homogeneous form; refer to the scale factor, w , in your answer. **[10 marks]**

$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k}$$

Explain the 2D form of Homogeneous coordinate described by the figure below. Explain where the points at infinity, (∞, ∞) , map to in Homogeneous space. Extend the diagram to show where parallel lines meet in homogeneous space.



(c) Describe three areas of the study of computer graphics where the features of the inner product or dot product play a significant role. **[10 marks]**

Continued next page.

- (d) Consider the OpenGL, GLSL shader code and run time display [10 marks]
shown and answer the questions that follow.

```

GLuint programID;
glm::mat4 MVP;
GLuint vertexbuffer;
GLuint uvbuffer;
GLfloat vertex_buffer_data[18] =
{-1,-1,0,1,-1,0,1,1,0,-1,-1,0,1,1,0,-1,1,0};
GLfloat uv_buffer_data[12] =
{0,0,1,0,1,1,0,0,1,1,0,1};
glm::mat4 projection;

int main(int argc, char* argv[])
{
    glutInitWindowPosition(200, 50);
    glutInitWindowSize(512, 512);
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE|GLUT_RGBA);
    glutCreateWindow("OpenGL");

    GLenum err = glewInit();
    glGenBuffers(1, &vertexbuffer);
    glGenBuffers(1, &uvbuffer);

    programID = LoadShaders("VS.vs", "FS.fs");
    Texture = LoadTextureBMP(".\\k.bmp");
    TextureID = glGetUniformLocation(programID, "text_samp");

    glm::mat4 projection = glm::ortho(-1,1,-1,1,-1,1);
    MVP = projection;

    glutDisplavFunc(Draw):

```

```

#version 330 core
in vec2 UV;
out vec4 color;

uniform sampler2D text_samp;

void main()
{
    color = texture(text_samp, UV).rgba;

```

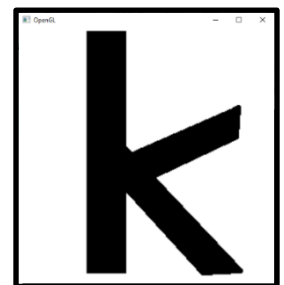
```

A { // if(color.r < 0.5)
    // {
    //   color.r=1;color.g=0;color.b=0;
    // }

B { // color.r=1-color.r;
    // color.g=1-color.g;
    // color.b=1-color.b;

C { // if(color.r < 0.5)
    // {
    //   color.r=UV.t;
    //   color.g=UV.t;
    //   color.b=UV.t;
    // }

```



- (i) Explain the term double buffering. Comment on how this is achieved using OpenGL.
- (ii) Name the shaders that are available to programmers of OpenGL. Which shader is shown above?
- (iii) Explain how the display changes when the commented code at "A", "B" and "C" are used separately.
- (iv) Create a labeled diagram of a likely set of polygons used to render the letter "k". Include reference to the vertex and UV coordinates. Explain your reasoning.
- (v) Describe the three transformations that are captured in the MVP matrix when OpenGL is being used to render 3D objects.

[40 marks]

- 3** (a) A Quaternion can be described using the following equation, [10 marks]

$$q = q_0 + iq_1 + jq_2 + kq_3$$

explain the terms contained in the equation. Evaluate q^2 in the case where $q_1=0$ and $q_3=0$.

In graphics the following form of quaternion is regularly used. Explain this form of the quaternion; include a diagram to support your answer.

$$r = \left(\cos\left(\frac{\theta}{2}\right), u \sin\left(\frac{\theta}{2}\right) \right)$$

Describe the quaternion operation that would be required to rotate a point $p=(0,p)$ about r , you do not need to include a full proof.

State two properties of Quaternion's that make them a popular choice for describing direction in computer graphics.

- (b) Describe how Euler angles can be used to describe the orientation of one frame with respect to another. In your answer mention the line of nodes. Describe the flaw with this approach to describing rotations. [10 marks]
- (c) Describe the algorithm used to create the logistic map. Sketch the resulting fractal and use it to explain the terms period doubling bifurcation and show how the fractal can be used to measure the Feigenbaum constant. [10 marks]

Continued next page.

- (d) Consider the blender python script and answer the questions that follow. [10 marks]

```

1 import bpy
2 import math
3
4 scene = bpy.context.scene
5
6 for obj in scene.objects:
7     bpy.context.collection.objects.unlink(obj)
8
9 bpy.ops.mesh.primitive_uv_sphere_add(location = (0,0,0), radius=1)
10 bpy.ops.object.shade_smooth()
11
12 bpy.ops.mesh.primitive_cone_add(location=(0,0,0),depth=2,radius1=0,radius2=1)
13
14 shape = bpy.context.active_object
15 mat = bpy.data.materials.new("mat")
16 mat.diffuse_color = (0, 0, 1, 1)
17 shape.data.materials.append(mat)
18
19 cam_data = bpy.data.cameras.new(name="cam")
20 cam_object = bpy.data.objects.new(name="Camera", object_data=cam_data)
21 bpy.context.collection.objects.link(cam_object)
22 scene.camera = cam_object
23
24 cam_object.location = (0, -8, 0)
25 cam_object.rotation_euler = (1.57,0,0)
26 cam = bpy.data.cameras[cam_data.name]
27 cam.lens = 30
28
29 scene.frame_start=0
30
31 positions = (-3,-3,0),(0,3,0),(3,-3,0),(-3,-3,0)
32 number_of_frame = 0
33 for position in positions:
34     scene.frame_set(number_of_frame)
35
36     shape.location = position
37     shape.keyframe_insert(data_path="location", index=-1)
38     scene.frame_end=number_of_frame
39     number_of_frame += 20

```

- (i) Describe the colour and position of the objects created in this script for the first frame of the animation.
- (ii) Describe the motion of the objects over the complete animation. How many frames are created? Explain your reasoning.

[40 marks]

- 4 (a) Explain how the depth sensor in the Kinect V1 or V2 works. [10 marks]
Include a diagram to support your answer. In your answer describe the physical construction of the device, the data it acquires and then comment on how the data collected is converted to a depth map.
- (b) A 3D object was created using the three files shown. Answer the [10 marks]
question that follow.

shape.obj

```
mtllib shape.mtl
o Shape
v -1 -1 0
v 1 -1 0
v 1 1 0
v -1 1 0
v 0 -1 1
v 0 1 1

vt 0 0
vt 1 0
vt 1 1
vt 0 1

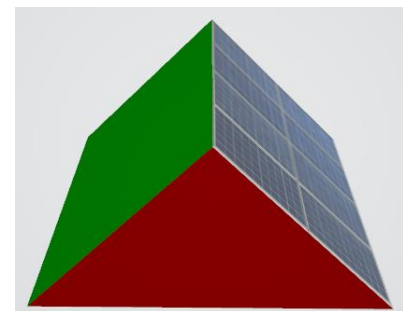
usemtl Mat
f 5/1 2/2 3/3 6/4

usemtl Green
f 1 2 3 4
f 1 5 6 4
```

shape.mtl

```
newmtl Mat
Ns 96.07
Ka 0.00 0.00 0.00
Kd 0.64 0.64 0.64
Ks 0.50 0.50 0.50
Ni 1.00
d 1.00
illum 2
map_Kd panel.bmp

newmtl Green
Ka 0.00 0.64 0.00
Kd 0.00 0.64 0.00
```

panel.mtl

- (i) Name the file format.
- (ii) Explain the terms vertex and texture coordinate.
- (iii) Explain how you would modify the code shown to add a red triangle at either end of the prism as shown.
- (c) Briefly explain each of the following colour models. In each case [10 marks]
explain the acronym and provide a brief explanation of how it captures or describes colour information.
- (i) RGB
- (ii) CMY
- (iii) LMS
- (iv) CIE
- (v) HSI

Continued next page.

- (d) Consider the following differential equations describing the motion of a and answer the questions that follow. [10 marks]

$$kv_y + m \frac{d^2 y}{dt^2} = 0$$

$$kv_x + m \frac{d^2 x}{dt^2} = 0$$

- (i) Explain the terms used.
- (ii) Rewrite one equation as two first order equations.
- (iii) Explain the term boundary condition.
- (iv) Write pseudocode (or any language) to show how the path of the projectile can be tracked over time.